Nationwide TDP Capstone Project 2026 V2.0

Sprint 4 — Final MVP, Security & Presentation

Duration: 3 Days

Objective

Integrate the complete solution, finish the MVP, validate the application end-to-end, strengthen security and prepare the final showcase.

Sprint Goal

A customer can use the completed solution to understand their **carbon impact and spending behaviour** and receive practical recommendations for reducing carbon and costs.

1. Agile & Team Working

- Review and revise the team's **Definition of Ready (DoR)** and **Definition of Done (DoD)**.
- Finalise the product and sprint backlogs.
- Maintain the task board and development log.
- Review team responsibilities and collaboration.
- Identify and resolve remaining blockers, defects and technical debt.

2. Complete the MVP

Integrate and validate:

- Customer spending capture.
- Spending categories.
- Emissions-factor data.
- Carbon calculations.
- Carbon-impact summaries.
- Spending/carbon trends.
- Practical recommendations.
- Potential cost-saving opportunities.

3. End-to-End Testing

Validate the complete customer journey:

Submit Spending

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Validate Data

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Store Data

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Retrieve Emissions Factor

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Calculate Carbon Impact

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Generate Insights

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Provide Recommendations

Testing should include:

- Unit testing.
- Integration testing.
- End-to-end testing.
- API testing.
- Data validation.
- Security testing.
- CI/CD validation.

4. DevSecOps

Complete the pipeline:



Focus on:

- Vulnerability remediation.
- Secure coding.
- Dependency security.
- Secrets management.
- Access control.
- Security testing.
- Automated quality/security gates.
- Evidence of security findings and remediation.

5. Responsible AI

AI may be used for:

- Code refinement.
- Test development.
- Troubleshooting.
- Documentation.
- Data analysis.
- Recommendation development.

Teams must **review, understand, test and security-check** AI-generated output.

The team should be able to explain where AI was used and demonstrate that generated content was appropriately validated.

6. Documentation

Complete the project documentation, including:

- Solution architecture.
- Technical decisions.
- API information.
- Database design.
- Testing approach.
- Security controls.
- DevSecOps pipeline.
- AI usage.
- Known limitations.
- Future improvements.

7. Final Showcase

Demonstrate the complete solution:

1. Submit customer spending.
2. Validate and securely store the data.
3. Retrieve emissions factors.
4. Calculate carbon impact.
5. Display carbon insights.
6. Show spending/carbon trends.
7. Provide recommendations.
8. Show potential cost savings.
9. Demonstrate automated tests.
10. Demonstrate the DevSecOps pipeline.
11. Show SAST, SCA and DAST evidence.
12. Demonstrate deployment.
13. Explain security decisions.
14. Explain how AI was used responsibly.

15. Explain the **customer and commercial value** of the solution.

Learning & Professional Outcomes

Sprint 4 provides evidence of:

- End-to-end software development.
- Testing and quality assurance.
- DevSecOps.
- Cybersecurity awareness.
- Responsible AI.
- Problem-solving.
- Teamwork.
- **Cross-disciplinary collaboration** across Development, Data, Security, Testing and Business.
- Communication.
- Commercial awareness.
- Ability to explain technical decisions in terms of customer value.

Sprint 4 Success

The team delivers a **working, tested and secure MVP** that demonstrates a complete customer journey from spending data through to carbon impact, insights, recommendations and potential savings, supported by CI/CD, security testing, documentation and a professional final showcase.